



Sass

Lecture 2



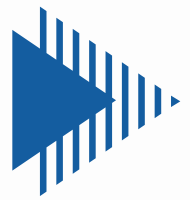
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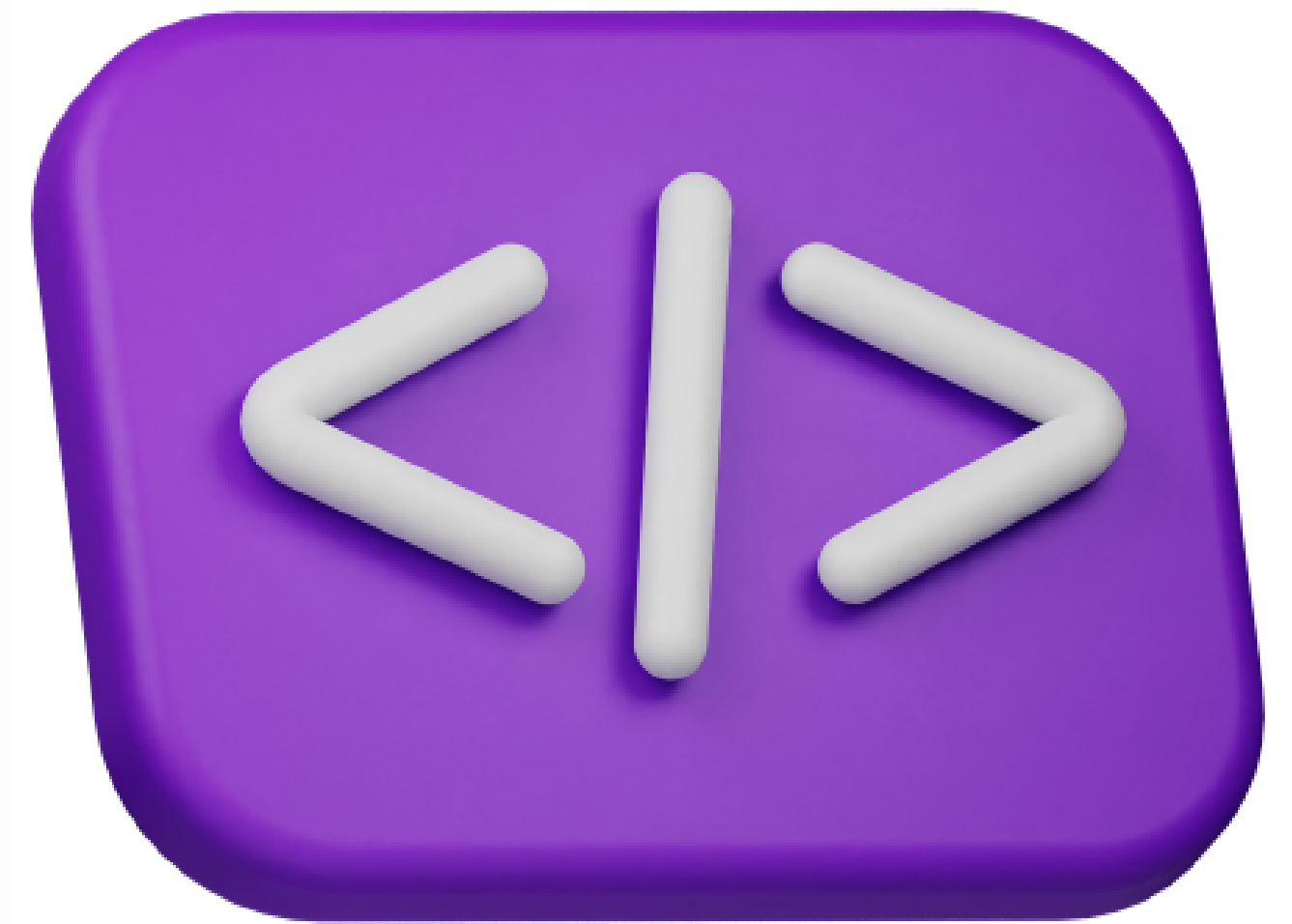
@for, @each , @while



Lists in Sass



Maps in Sass



Sass provides several control directives that allow us to write more efficient and reusable styles:

- @for** → Loops through a range of numbers.
- @each** → Iterates over lists and maps
- @while** → Loops while a condition is true.
- Lists** → Ordered collections of values.
- Maps** → Key-value pairs for structured data.



@for Loop

The **@for** directive loops over a range of numbers and allows you to **dynamically** generate styles based on a counter variable.

Syntax:

A screenshot of a code editor window with a dark theme. The title bar shows three colored circles (red, yellow, green) and the text "Sass for.scss". The code content is as follows:

```
@for $i from <start> through <end> {  
    // Code to execute  
}
```

Example:



```
for.css

.box-1 { width:10px; height:10px; }
.box-2 { width:20px; height:20px; }
.box-3 { width:30px; height:30px; }
.box-4 { width:40px; height:40px; }
.box-5 { width:50px; height:50px; }
```



```
for.scss

@for $i from 1 through 5 {
  .box-#{ $i } {
    width: #{ $i * 10 }px;
    height: #{ $i * 10 }px;
  }
}
```

@each Loop

The **@each** directive is used to loop over lists and maps, making it useful for creating CSS classes dynamically.

Syntax:



```
@each $variable in <list> {  
  // Code block  
}
```

Lists in **Sass**

A **list** is an ordered collection of values, separated by commas or spaces. Lists help store **multiple values** efficiently like **array**.

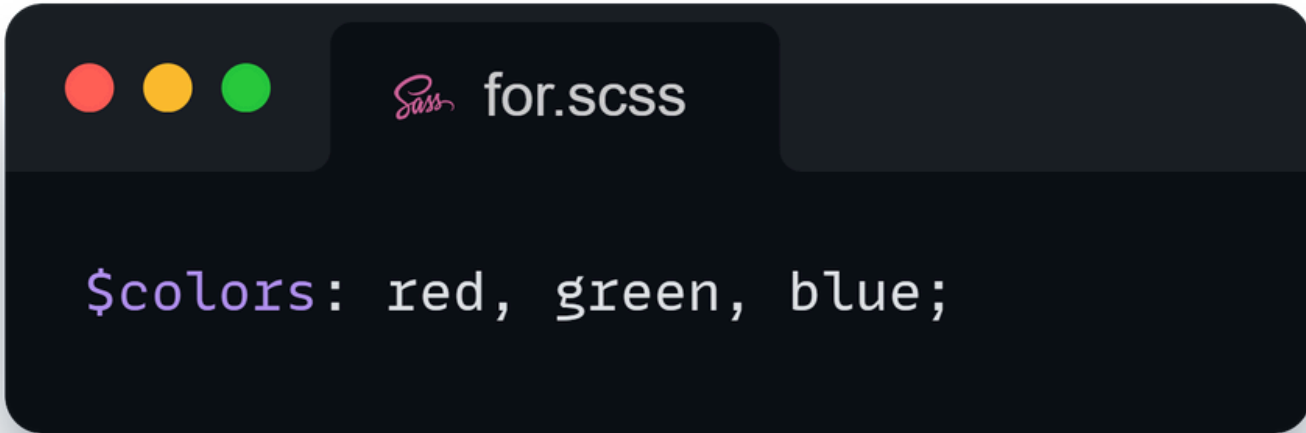
Syntax:

A code editor window with a dark theme, titled 'for.scss'. It contains the following Sass list syntax:

```
$list-name: item1, item2, item3;
```

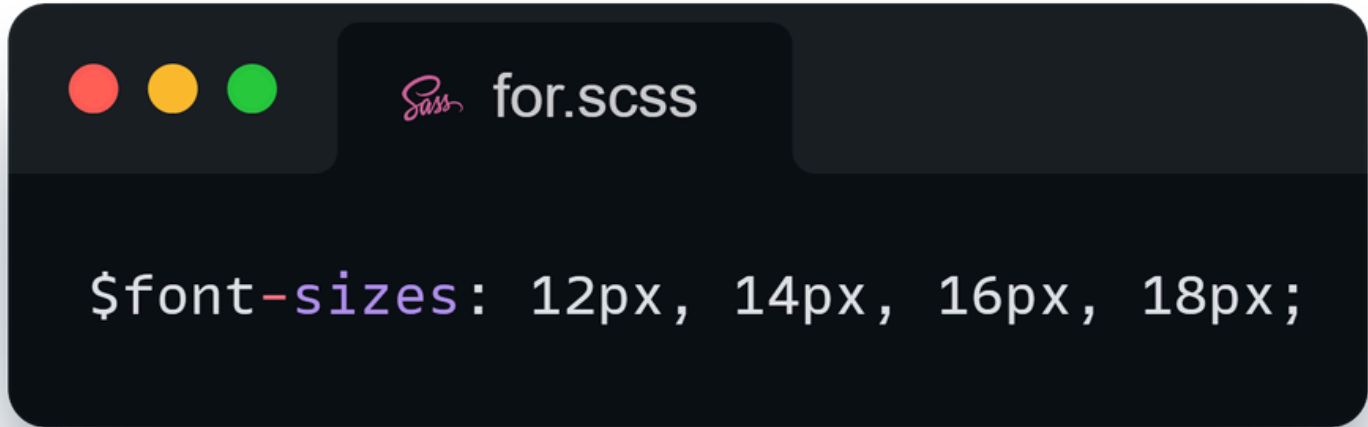
```
$list-name: item1, item2, item3;
```

Examples:

A code editor window with a dark theme, titled 'for.scss'. It contains the following Sass list syntax:

```
$colors: red, green, blue;
```

```
$colors: red, green, blue;
```

A code editor window with a dark theme, titled 'for.scss'. It contains the following Sass list syntax:

```
$font-sizes: 12px, 14px, 16px, 18px;
```

```
$font-sizes: 12px, 14px, 16px, 18px;
```

Example of @each with List



```
for.css

.text-red { color: red; }
.text-green { color: green; }
.text-blue { color: blue; }
.text-yellow { color: yellow; }
```

```
for.scss

$colors: red, green, blue, yellow;

@each $color in $colors {
  .text-#{ $color } {
    color: $color;
  }
}
```

Map in Sass

A **map** is a collection of **key-value pairs**. Maps are useful for storing structured data.

Syntax:

A dark-themed code editor window with a title bar containing three colored circles (red, yellow, green) and the text "Sass for.scss". The code inside is:

```
$map-name: (  
  key1: value1,  
  key2: value2  
);
```

```
$map-name: (  
  key1: value1,  
  key2: value2  
);
```

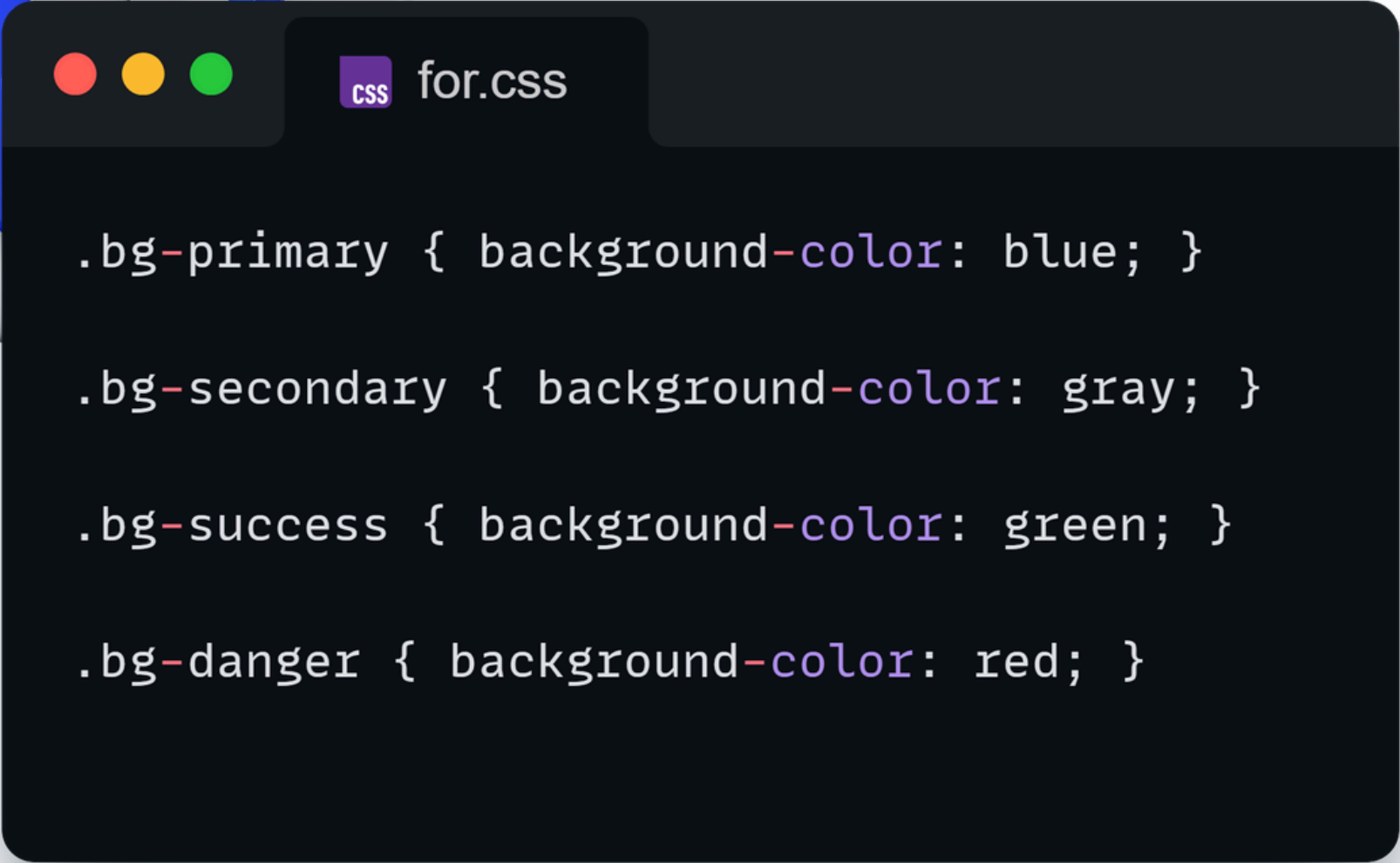
Examples:

A dark-themed code editor window with a title bar containing three colored circles (red, yellow, green) and the text "Sass for.scss". The code inside is:

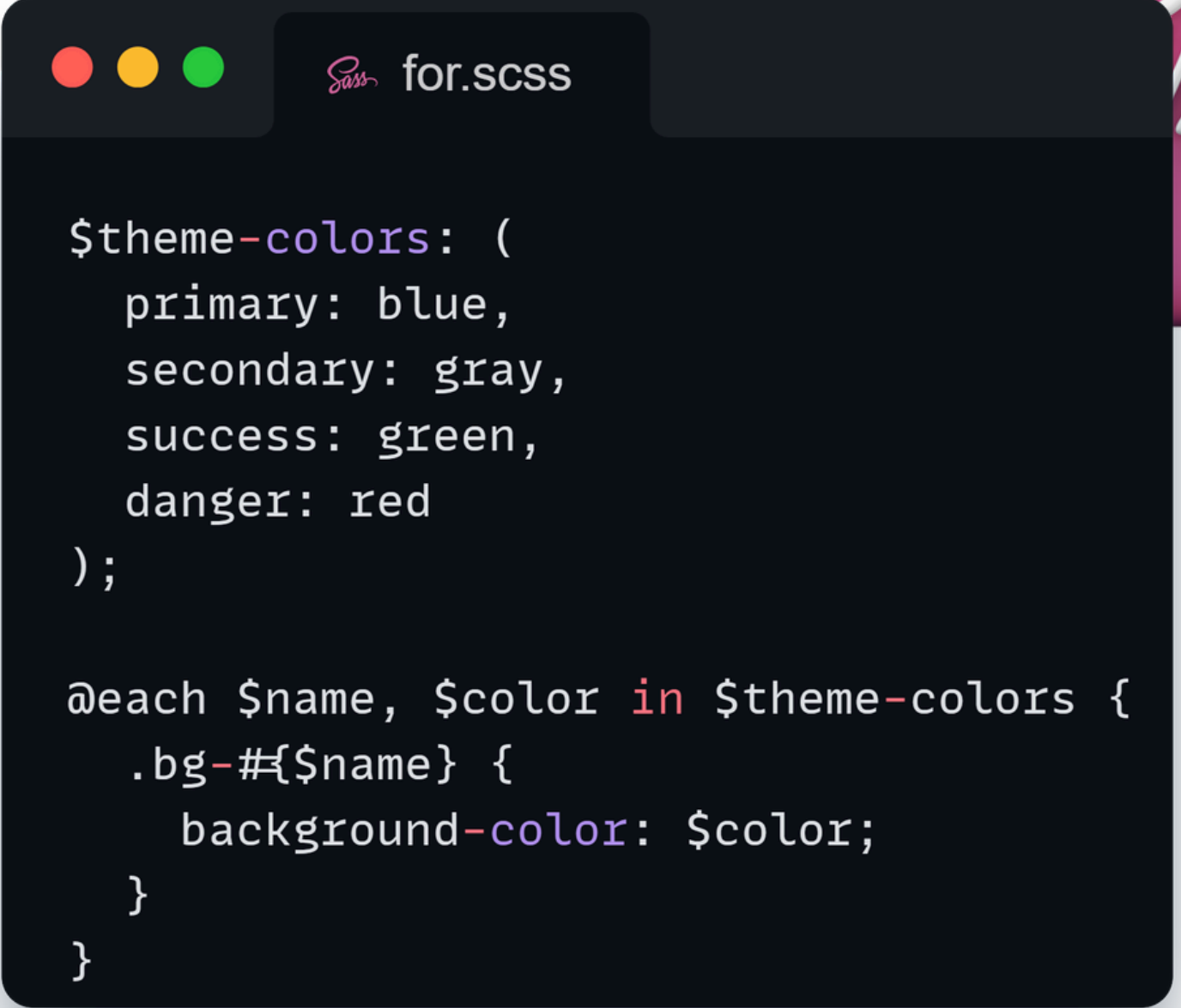
```
$spacing: (  
  small: 5px,  
  medium: 10px,  
  large: 15px  
);
```

```
$spacing: (  
  small: 5px,  
  medium: 10px,  
  large: 15px  
);
```

Example of @each with Map

A dark-themed code editor window titled 'for.css' with a CSS icon. It contains four CSS rules for background colors. The window is positioned in front of a white folder icon with blue tabs.

```
.bg-primary { background-color: blue; }  
.bg-secondary { background-color: gray; }  
.bg-success { background-color: green; }  
.bg-danger { background-color: red; }
```

A dark-themed code editor window titled 'for.scss' with a Sass icon. It contains a map of theme colors and an @each loop that generates CSS rules. The window is positioned in front of a pink folder icon with a white squiggle.

```
$theme-colors: (  
  primary: blue,  
  secondary: gray,  
  success: green,  
  danger: red  
);  
  
@each $name, $color in $theme-colors {  
  .bg-#{ $name } {  
    background-color: $color;  
  }  
}
```

Sass

```
for.scss

// 2. Text colors using @each and a list
$colors: red, blue, green;
@each $color in $colors {
  .text-#{ $color } {
    color: $color;
  }
}

// 3. Button styles using @each and a map
$btn-colors: (primary: blue, danger: red);
@each $name, $color in $btn-colors {
  .btn-#{ $name } {
    background-color: $color;
    color: white;
    padding: 8px 16px;
    border: none;
    border-radius: 4px;
  }
}
```

HTML

```
sass.html

<body>

  <h3>Boxes (@for loop)</h3>
  <div class="box-1"></div>
  <div class="box-2"></div>
  <div class="box-3"></div>

  <h3>Text Colors (@each loop)</h3>
  <p class="text-red">Red Text</p>
  <p class="text-blue">Blue Text</p>
  <p class="text-green">Green Text</p>

  <h3>Buttons (@each & map)</h3>
  <button class="btn-primary">Primary</button>
  <button class="btn-danger">Danger</button>

</body>
```

THANK YOU!

Diamonds are born under pressure.
If you can handle this, you will
become a diamond.

